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Lab 6

Part 1 – Recap: Lighting and Shadows in Shadows.py

1.1. Reading and understanding

1. Lights in the Scene

Q1. Ambient Light

```
# --- Lighting ---  
self.ambient_light = AmbientLight(color=[0.3, 0.3, 0.3])  
self.scene.add(self.ambient_light)
```

- a. It is the Dark Gray color.
- b. The role of ambient light is providing a base level of illumination that evenly lights all objects in a scene, preventing them from appearing completely dark, while directional light simulates sunlight by coming from a specific direction, creating shadows, highlights, and depth. When combined, ambient light softens harsh contrast and directional light adds realistic shading, resulting in a balanced and natural-looking scene.

Q2. Directional light (The Sun)

```
self.sun_light = DirectionalLight(color=[1.0, 1.0, 0.9], direction=[-1, -1, -1])  
self.scene.add(self.sun_light)
```

- a. It is the warm white color.
- b. The directional light is initially given the vector $[-1, -1, -1]$, meaning the sunlight is coming diagonally from above and behind the scene, pointing downward in the negative x, y, and z directions.
- c. A directional light is a good model for sunlight because the Sun is extremely far away, so the light rays that reach Earth are effectively parallel, producing consistent lighting and shadows that do not depend

on distance—just like how directional lights illuminate objects based only on light direction, not on the light's position.

Q3. The difference between the Ambient Light and the Directional Light.

- **Ambient light** has **no direction**, so it illuminates all surfaces equally and **does not depend on the surface normal**; every point receives the same base brightness.
- **Directional light**, on the other hand, **has a specific direction**, and its intensity **depends on how the surface normal aligns with that direction** (brighter when the surface faces the light, darker when angled away).

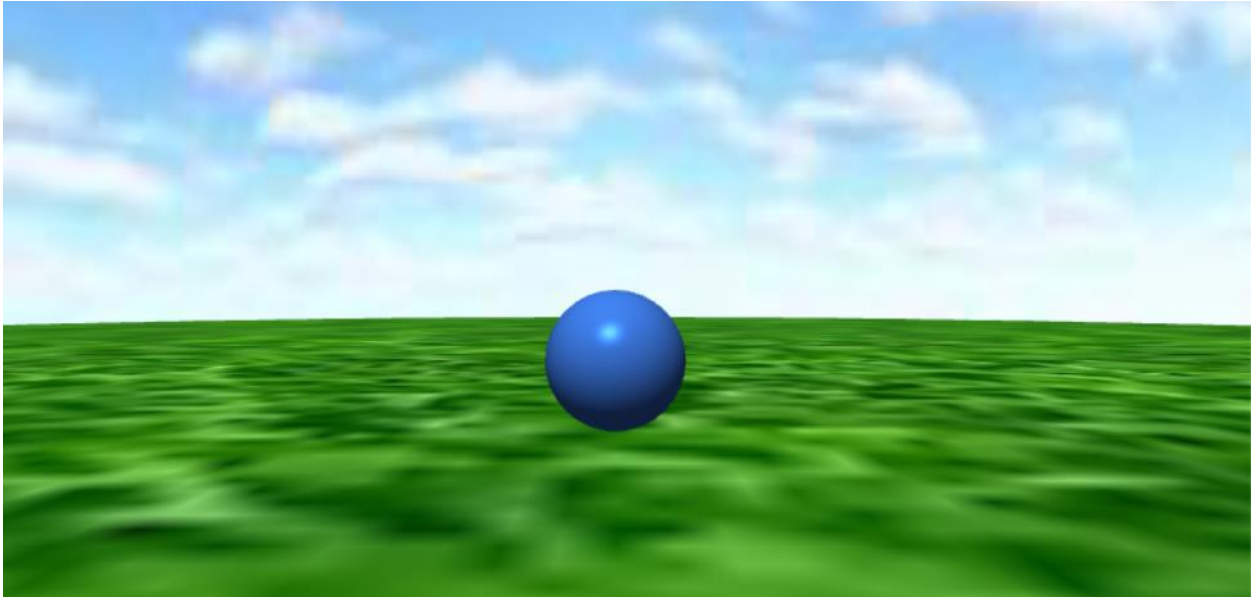
2. Materials and Shadows

Q4.

```
# --- Earth sphere (shaded by lights + casts shadow) ---
earth_geometry = SphereGeometry(radius=1.0)
earth_material = PhongMaterial(
    property_dict={"baseColor": [0.2, 0.4, 0.8]},
    number_of_light_sources=2,    # ambient + directional
    use_shadow=True              # enable shadow
)
self.earth_object = Mesh(earth_geometry, earth_material)
self.earth_object.set_position([0, 1, 0])
self.scene.add(self.earth_object)
```

```
# --- Grass (Phong so it can receive shadows) ---
grass_geometry = RectangleGeometry(width=100, height=100)
grass_material = PhongMaterial(
    texture=Texture(file_name="textures/grass.jpg"),
    number_of_light_sources=2,
    use_shadow=True
)
grass = Mesh(grass_geometry, grass_material)
grass.rotate_x(-math.pi / 2)
self.scene.add(grass)
```

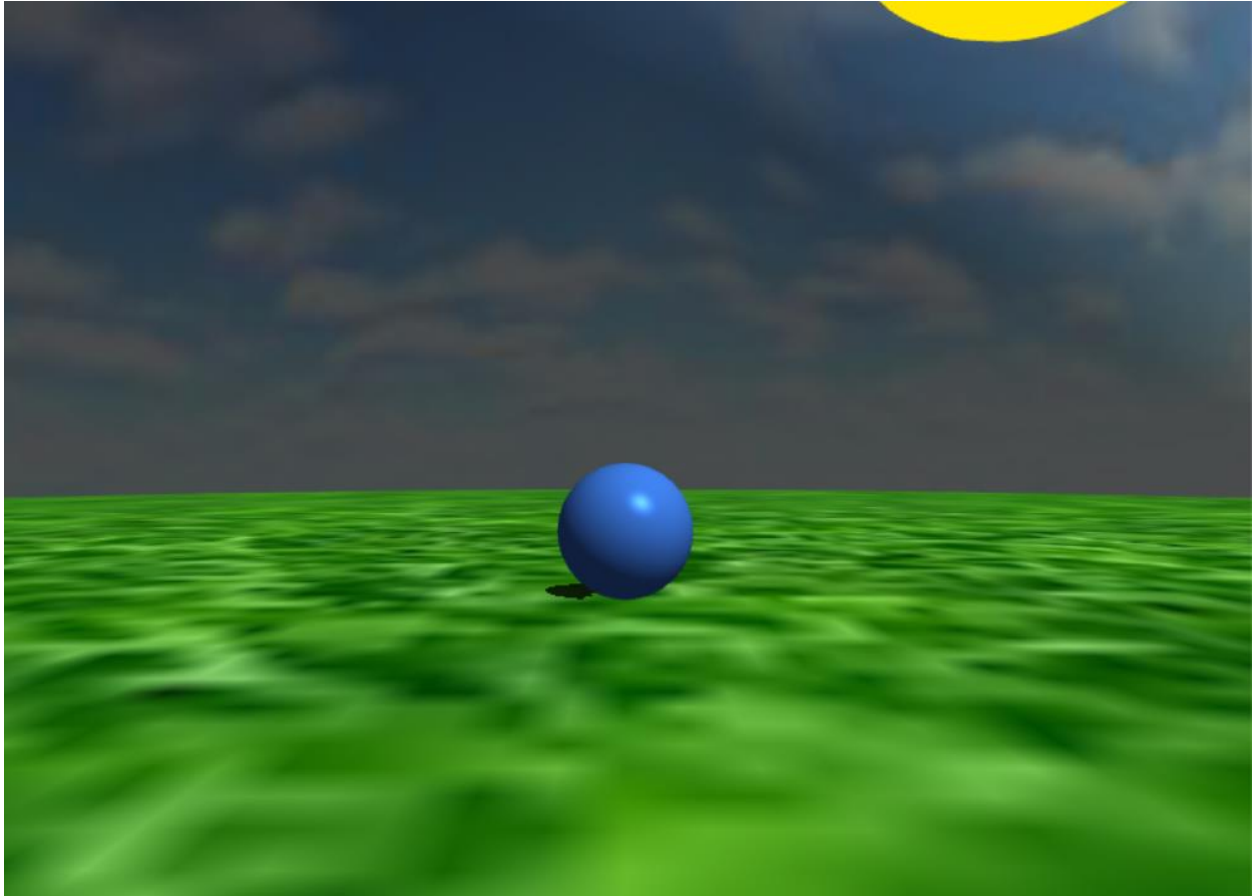
- a. It is **number_of_light_sources** and **use_shadow**.
- b. If `use_shadow` is set to `False` for the ground, it will not appear. For example we set the `False` of `use_shadow` in the `grass_material`.



Q5.

```
1 self.sun_light = DirectionalLight(color=[1.0, 1.0, 0.9], direction=[-1, -1, -1])
2 self.scene.add(self.sun_light)
```

- a. Because in real life, the object is not interacting with the sky, like the ground, so when the sun shines to the object, the shadow won't appear.
- b. It will create a shadow behind the object in every time the sun moves toward the object.



Q6. Shadow mapping

```
# Enable shadows from the Sun  
self.renderer.enable_shadows(  
    self.sun_light,  
    strength = 0.8  
)
```

- a. The sun_light
- b. The parameter strength controls the sun light is power. On the other words, if the strength increases, the opacity of the shadow increases.

3. Sun Movement and Shadow Direction

Q7.

```
# Animate the Sun across the sky
sun_x = 6 * math.cos(0.3 * self.time)
sun_y = 8
sun_z = 6 * math.sin(0.3 * self.time)
self.sun.set_position([sun_x, sun_y, sun_z])
```

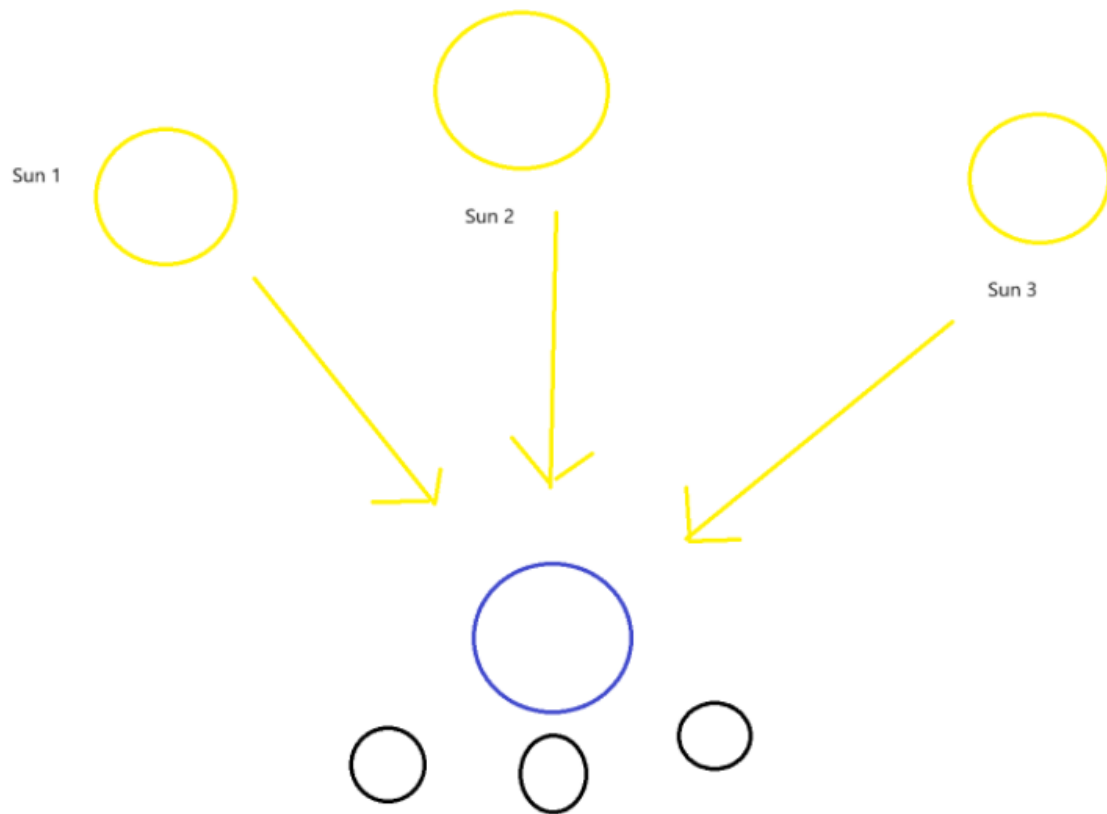
- a. It is math.cos and math.sin
- b. It is the “0.3 * self.time) as velocity.

Q8.

```
116
117 # Light direction follows the Sun sphere position
118 target = [0, 1, 0]
119 dir_x = target[0] - sun_x
120 dir_y = target[1] - sun_y
121 dir_z = target[2] - sun_z
122 self.sun_light.set_direction([dir_x, dir_y, dir_z])
123
```

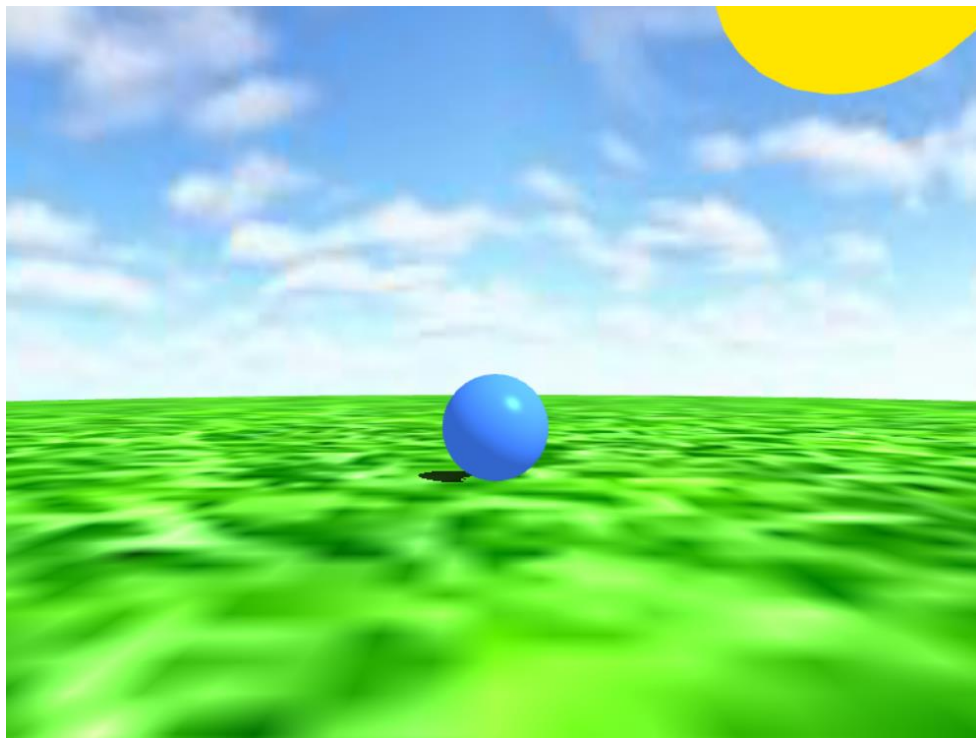
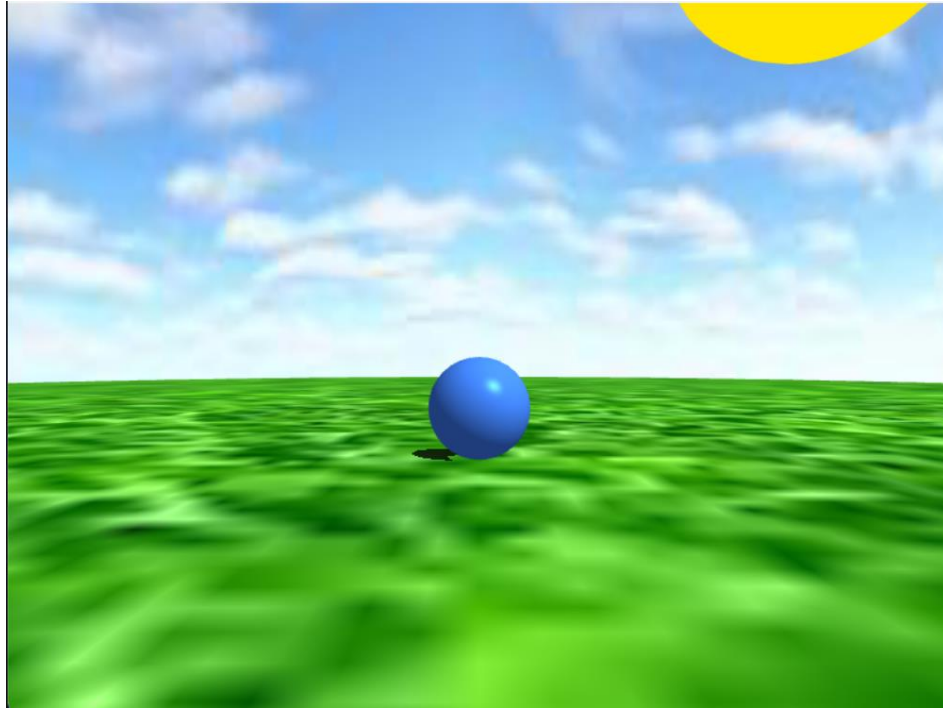
- a. The “target” point is the point the light is aimed at, and in the code, it is set to [0, 1, 0]. The direction of the light is computed by subtracting the Sun’s current position from this target point, giving a vector that points from the Sun toward the target.
- b. Changing the target point changes the direction that the directional light is aiming toward; since shadows are cast opposite the light direction, modifying the target would make the shadows tilt or shift in a different direction. For example, aiming the light at a different point on the ground would cause shadows to rotate or change their angle across the scene.

Q9.



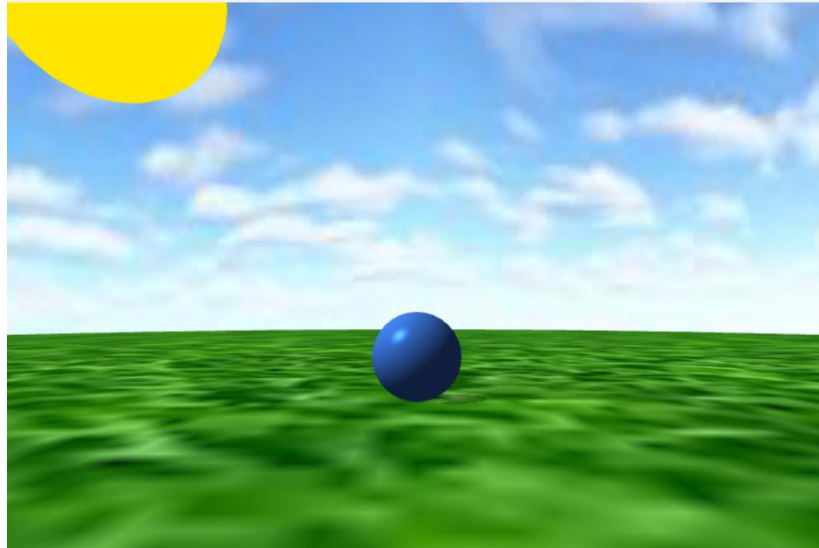
1.2. Playground: “What If” Experiments

P1. Ambient intensity:

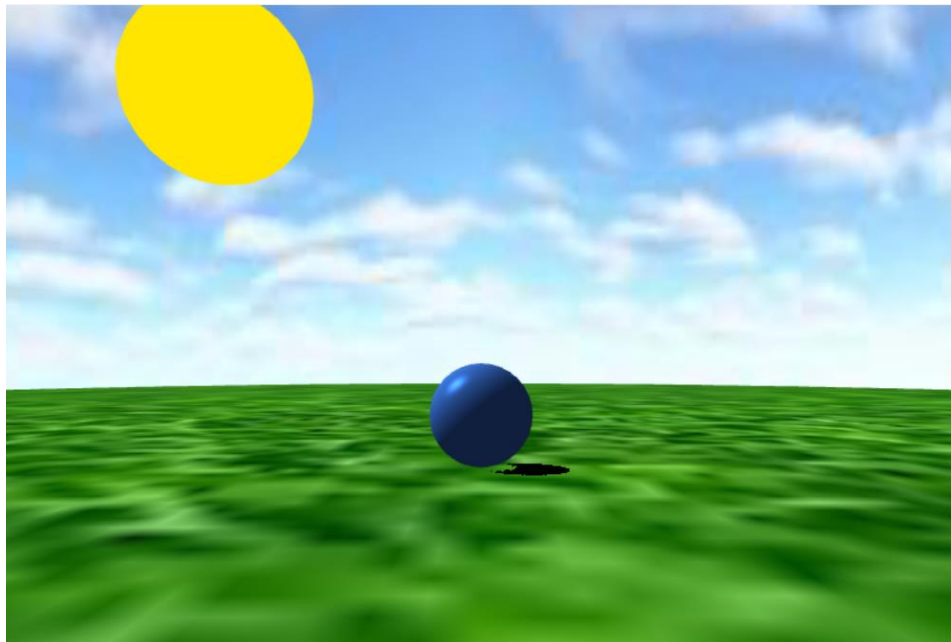


The shadows and darks area look darker. It might because the light illuminates.

P2. Shadow strength:



For the strength = 0.3, the shadow visibility seems to be fading away.



For the strength = 0.98, the shadow visibility is more clearly.

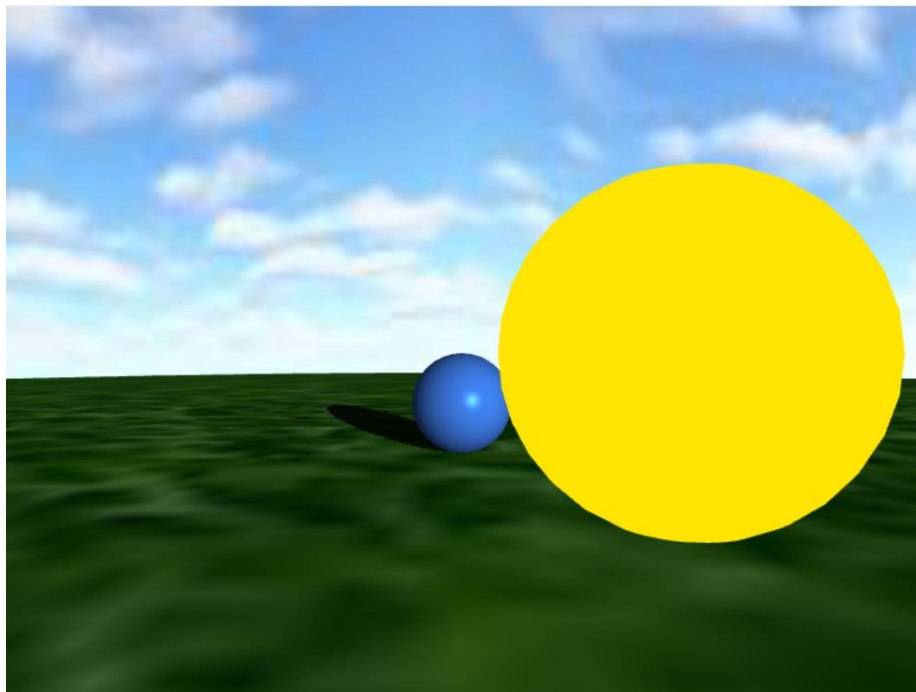
P3. Disabling ground shadow:

- a. When `use_shadow=False` is set for the ground plane, the Earth sphere still casts a shadow according to the light, but the ground no longer

receives that shadow. The Earth itself remains shaded correctly (lit by the Sun), but the dark shadow on the ground disappears.

- b. This shows that lighting and shadows are separate processes. A surface can be lit (affected by light direction, color, and material) without receiving shadows. Being lit means the surface's brightness depends on the light and its orientation. Receiving a shadow is an extra step where other objects block the light and darken parts of that surface. Removing the ground's ability to receive shadows proves it is still lit normally, but simply ignores blocked light coming from other objects.

P4. Sun height:



- a. The shadow lengths look taller when the Sun is lower.
- b. Long shadows at sunrise and sunset occur because the Sun is low on the horizon, causing its light to strike objects at a shallow angle. This makes the shadows stretch far across the ground, whereas when the Sun is overhead at midday, light hits surfaces more directly, producing short shadows. Thus, the lower the Sun in the sky, the longer the shadows appear.

P5. Sun speed:

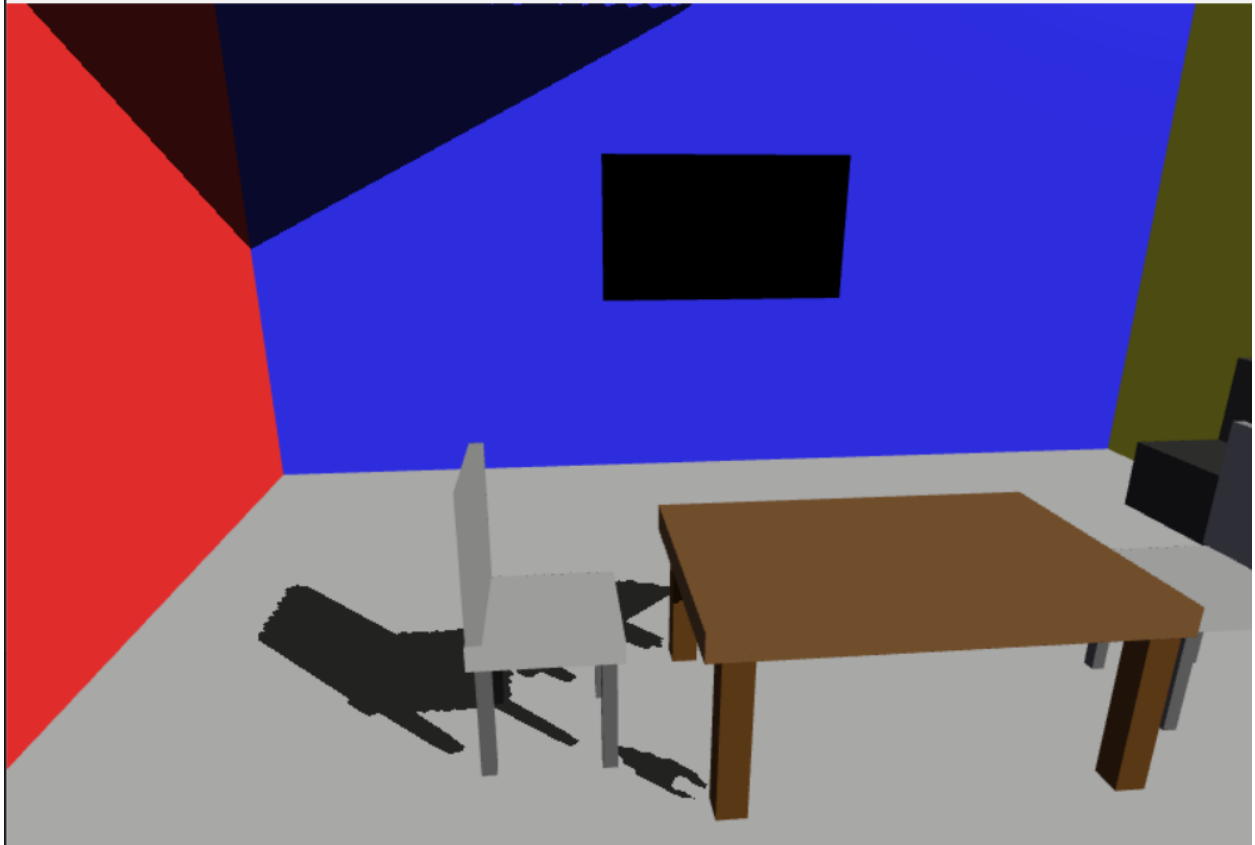
- a. It moves faster but still moves around the Earth.
- b. In your animation, the Sun moves noticeably across the sky while the Earth rotates slowly. Since a full day-night cycle in reality takes 24 hours, and your Sun completes its arc in a few seconds of animation, the time-lapse roughly compresses an entire day into seconds or minutes. So the animation represents a 24-hour physical time scale condensed into a very short visual sequence.

Part 2 – Applying Lighting and Shadows to ex4_7.py

Task 1

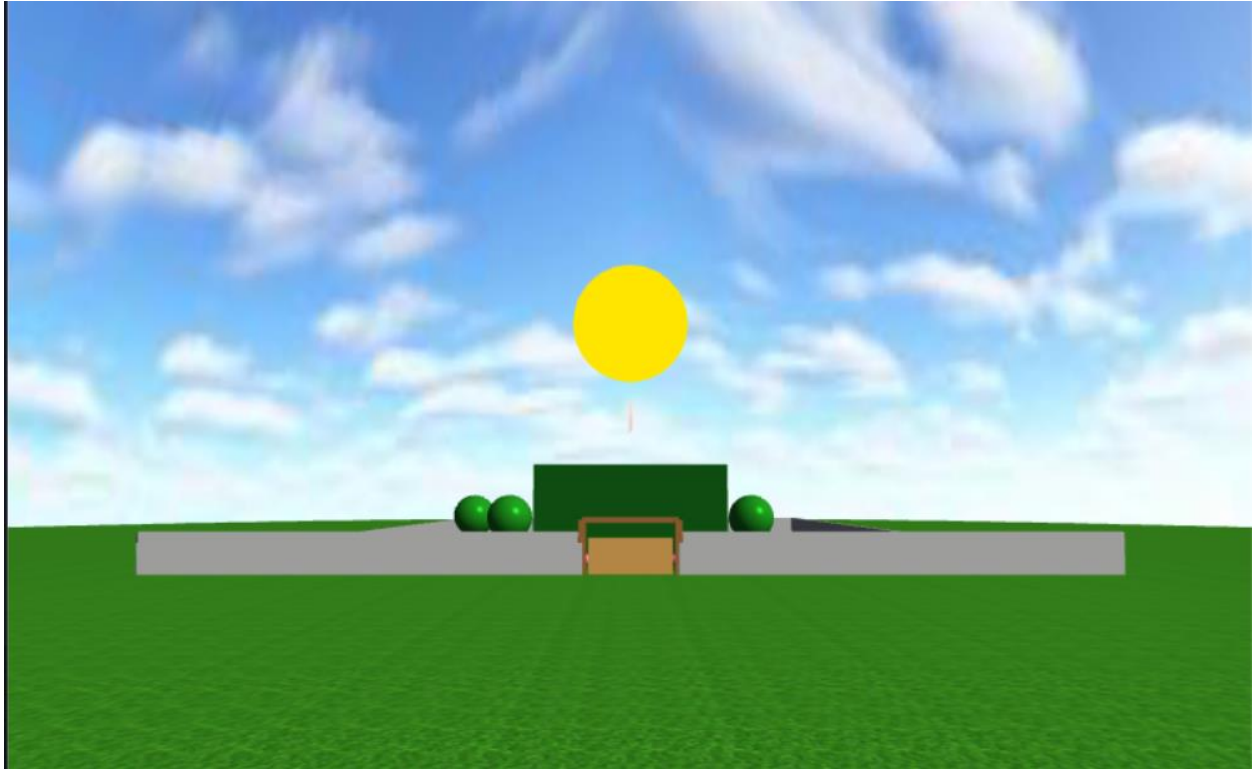
- If you comment out (disable) the ambient light: The scene will have strong directional lighting only, so areas not directly lit by the directional light (shadows, or surfaces facing away from the light) will appear almost completely black. The overall scene looks harsh and lacks fill light.
- If you comment out the directional light and keep only ambient light: The scene will appear evenly lit, flat, and without any shading or shadows. Objects lose depth cues, highlights, and contrast, making the scene look “cartoon-like” or flat.

Task 2



- **Three objects in the scene that should clearly receive shadows**
 - Floor / Ground plane
 - Trees
 - Furniture (tables, chairs, sofa)
- **One object that realistically does not need to receive shadows:** Sky dome / sky sphere — it is far away and uniform, so no shadows are needed

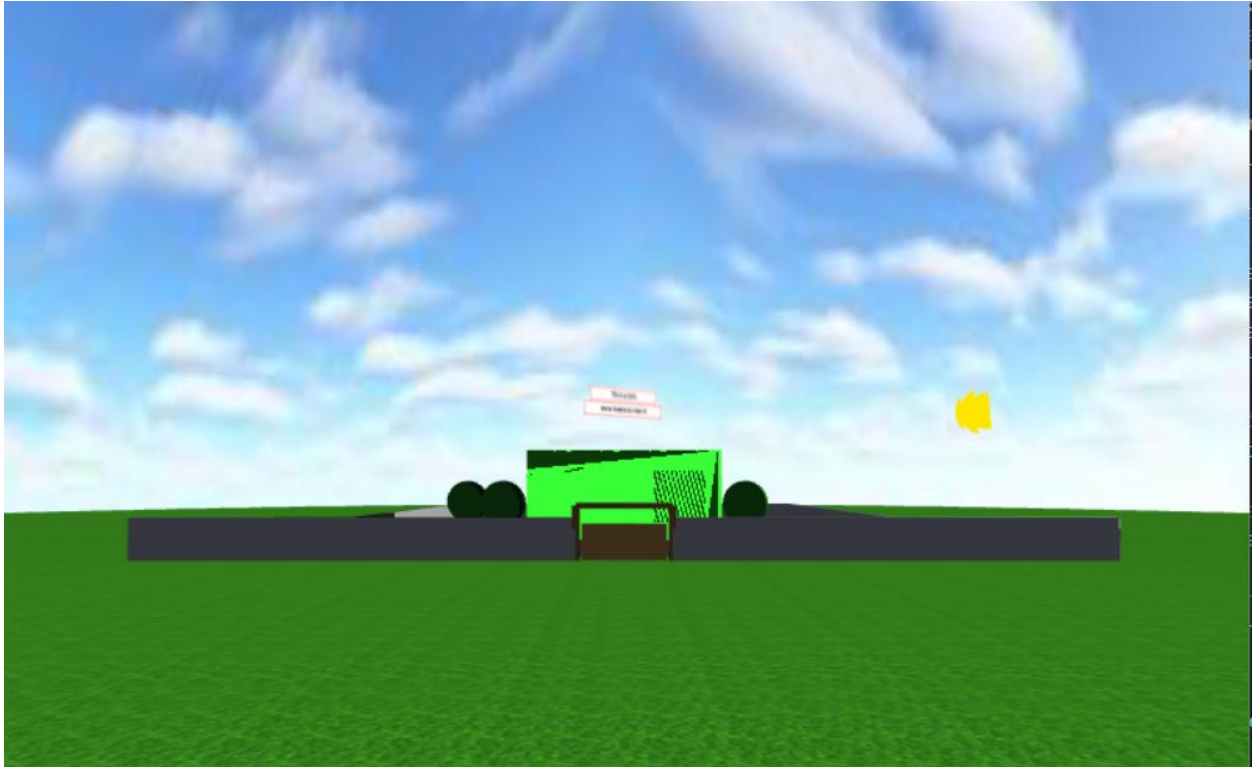
Task 3



Why is it useful to render the Sun as a simple colored sphere instead of a Phong-lit object?

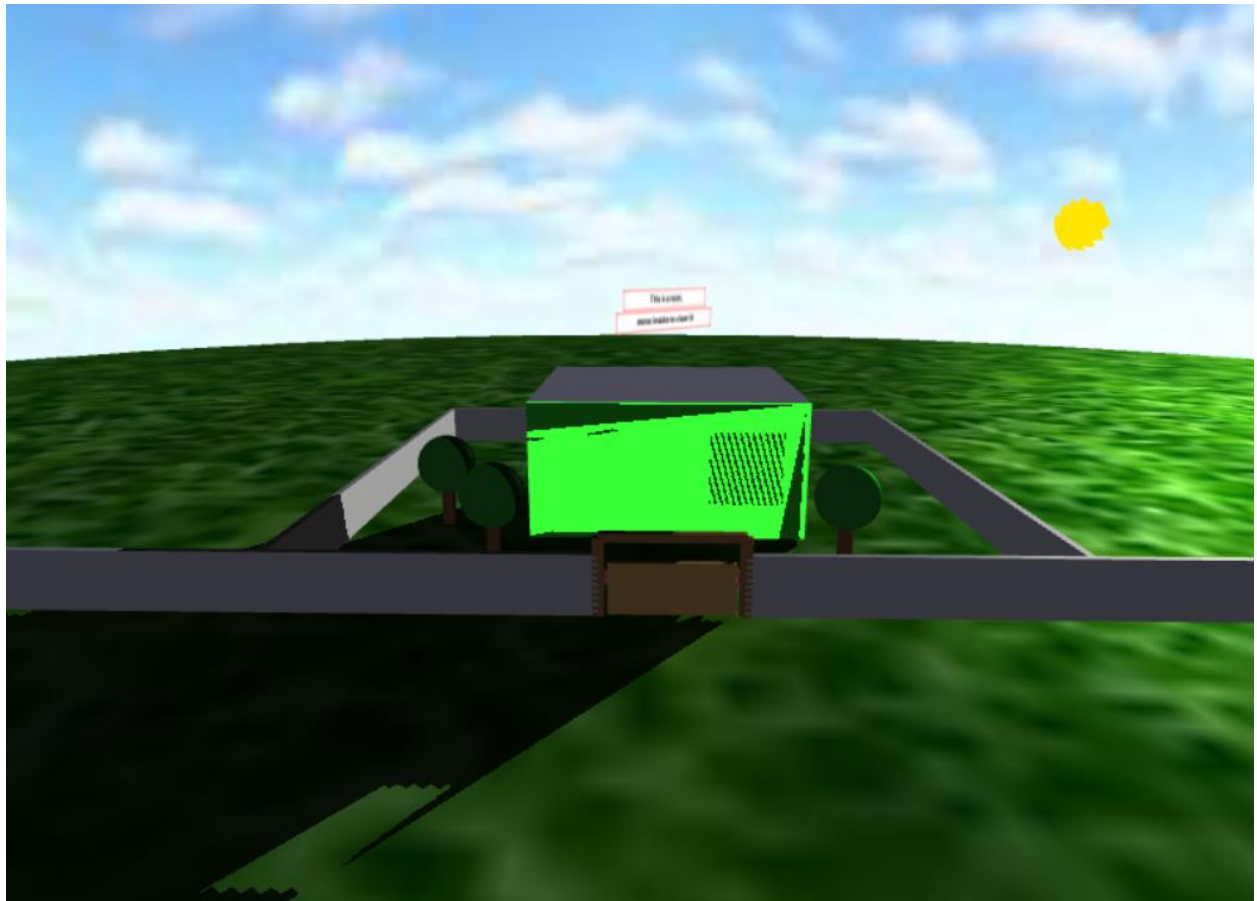
- ➔ Rendering the Sun as a simple unlit yellow sphere ensures it appears consistently bright regardless of other lights or shading in the scene. Using a Phong-lit material would cause the Sun to be affected by shadows and directional lighting, making it look dim or unrealistic, which is not physically accurate for the Sun.

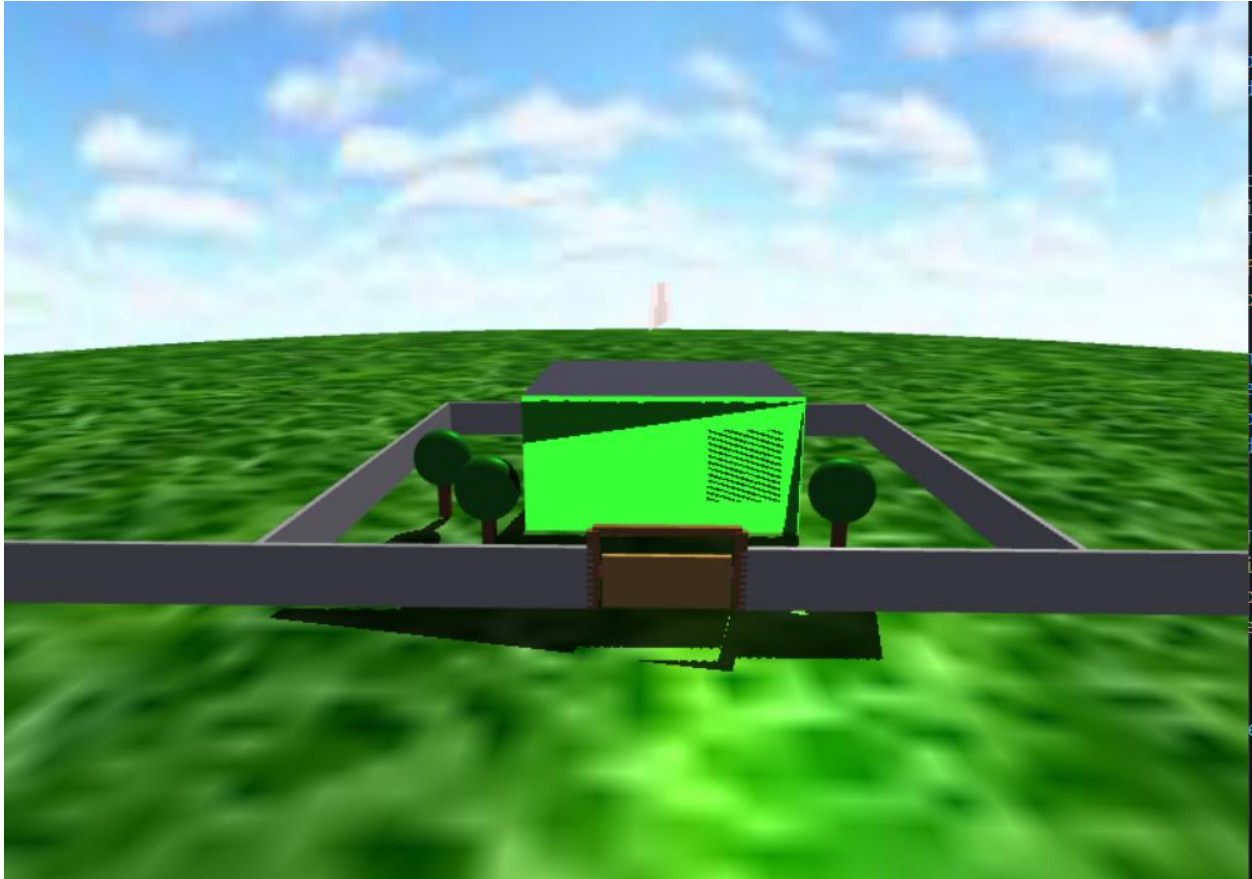
Task 4



- Shadows of the fence, trees, and other objects rotate and slide across the ground from east to west, following the Sun's motion.
- **Longest shadows:** Occur early morning and late afternoon, when the Sun is near the horizon (low sun_y).
- **Shortest shadows:** Occur around midday, when the Sun is high in the sky (maximum sun_y), casting nearly vertical shadows.

Task 5





- If ambient light is too strong relative to directional light, shadows become washed out, the scene looks flat, and the sense of depth is reduced.
- Best results occur when ambient fills just enough light to see shadowed areas, while directional light creates contrast and strong shadow cues.

Part 3—Extending Your Lab5 Scene with Lighting and Shadows

Task 6

